

# University of Sri Jayewardenepura

## Department of Mathematics

MAT 211 1.0

Linear Algebra II

Semester I 2026

### Assignment 2

**Due: June 5<sup>th</sup>, 2026 by 12 noon**

Let  $A$  and  $B$  be square matrices of the same order. We say that  $B$  is similar to  $A$  if there exists an invertible matrix  $P$  such that  $B = P^{-1}AP$ .

Prove the following.

1. The relation *similarity* is an equivalence relation.
2.  $A$  and  $P^{-1}AP$  have the same determinant.
3.  $A$  is invertible if and only if  $P^{-1}AP$  is invertible.
4.  $A$  and  $P^{-1}AP$  have the same characteristic polynomial
5.  $A$  and  $P^{-1}AP$  have the same eigenvalues.



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**Reference:** Elementary Linear Algebra by Grossman.